

# Get better inside the groove.

Pocket / A rehearsal room with a convincing backing band, where musical challenges and take reviews turn practice into playing.

DESIGN PROPOSAL · NOT A BUILT PRODUCT

The unit of progress is a musical role performed reliably in context.

For guitarists and bassists motivated by groove, improvisation and performing with others. Imported pieces remain first-class rehearsal material.

Open a piece or describe a jam. Choose your role, play with a great-sounding band, and get one specific way to sound more together.

Pocket

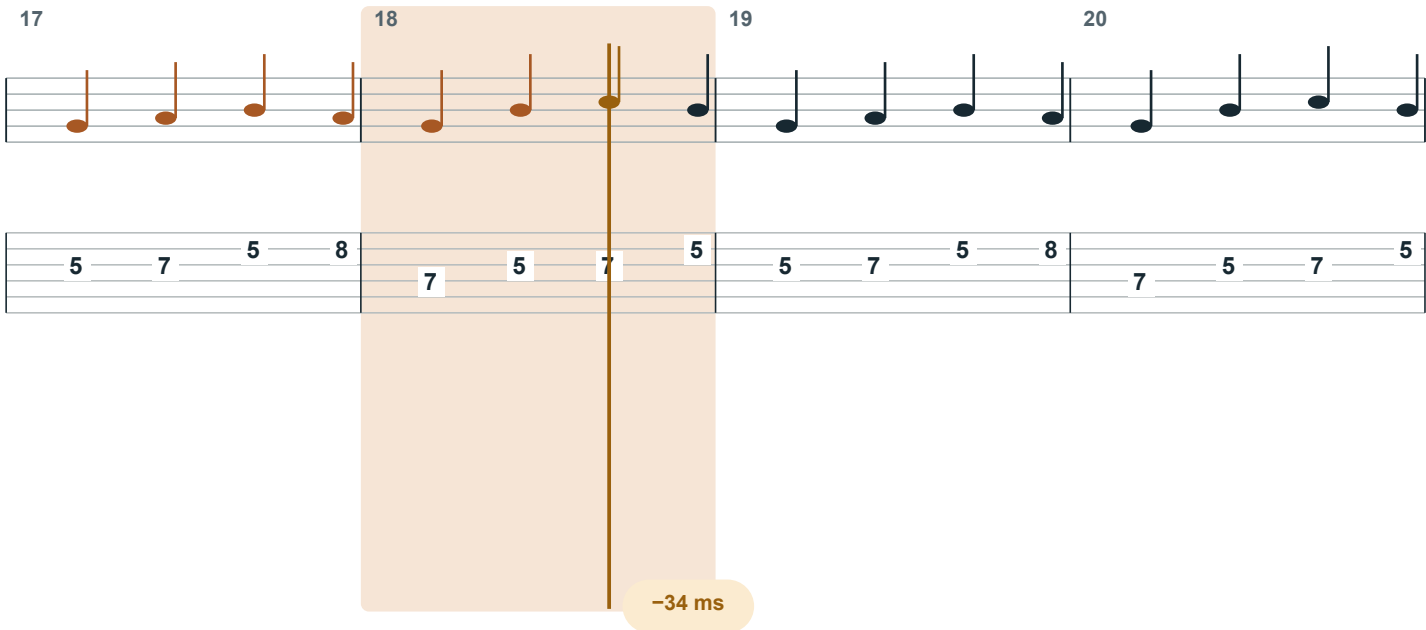
LibraryPracticeJamProgress

Input: ready

Rehearsal / Evening Study

72 BPM

17181920



Play loop

Bars 17–20 • Count-in 1 bar

Feel

The band sounds good enough to make another chorus inviting.

Context

Challenges measure a stated musical role, not conformity to one improvised solo.

Momentum

Feedback arrives between passes so the music remains the focus.

# A complete practice app, with one clear center

Rehearse → Piece or Jam → Take review; Set list, Mixer and Progress are persistent.

## Rehearsal room

A set list combines imported pieces, loops and jam recipes. The score remains the main playing surface.

## Band builder

Editable chord form, style, swing, key, role and density. Preview the result before playing; mute the user’s role.

## Musical challenges

Click gaps, call-and-response, roots at changes and written groove patterns. Each challenge names exactly what is assessed.

## Rehearsal cards

Save the arrangement seed, sound pack, tempo, role and evidence. Compare takes on equivalent tasks.

### COMMON CAPABILITIES IN THE FIRST COMPLETE RELEASE

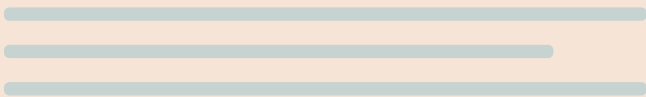
| Bring music                                                                 | Hear and control                                                       | Play and improve                                                                     | Keep and return                                                                     |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| GP3–8 + MusicXML import; track and tuning confirmation; score / TAB.        | Sampled band; track mute/solo; 40–200% speed; A–B loops; count-in.     | Mic / interface input; supported note and timing assessment; evidence-linked drills. | Per-phrase tempo history; reliable-score coverage; saved exercises and jam recipes. |
| Import report lists degraded or unsupported features. Keep the source file. | Learn mode, assessment mode, mixer, metronome and reference listening. | Clear recording-confidence states; replay; report an incorrect observation.          | Guest practice with local saves; optional account sync and deletion controls.       |

# A practice session with a beginning and an outcome

Six frames follow one guitarist or bassist from choosing music to returning with a reason. All interface data are illustrative.

01 Choose music or a jam

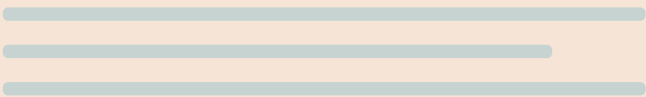
Import a part, or describe a groove



Riley imports a Guitar Pro arrangement and selects bass. Another day can begin with “shuffle beat ii–V–I in A.”

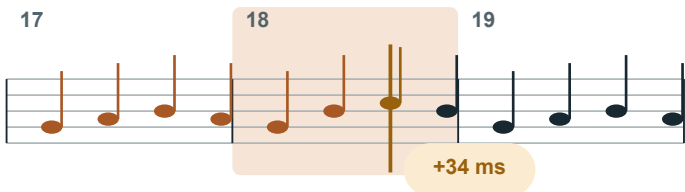
02 Meet the band

Mute the part you will play



Preview each instrument, choose the groove density, and check input balance. The band shows the exact form before starting.

03 Play a chorus



A Guitar Pro-like score view follows the band. Riley loops a transition and drops from 90 to 72 BPM.

04 Hear the useful difference

A specific observation

Early after the quiet bar.

A take review highlights the entry and lets Riley hear it against the band. The app suggests one groove challenge.

05 Trade a phrase



A generated call-and-response exercise uses the score’s rhythm. The backing parts stay fixed during an assessed pass.

06 Keep a rehearsal card



Riley saves the form, sound, tempo and challenge result. A later rehearsal tests the same task and then a new key.

# The score remains the place where playing happens

Desktop-first concept. Vector mockup uses an invented phrase and illustrative data; notation is schematic.

Pocket

LibraryPracticeJamProgress

Input: ready

Set list

01 Sound check

Input + band balance

02 Learn the part

Loop bars 17–20

03 Hold the groove

Jam in A · shuffle

04 Full take

90 BPM · review after

Rehearsal / Evening Study

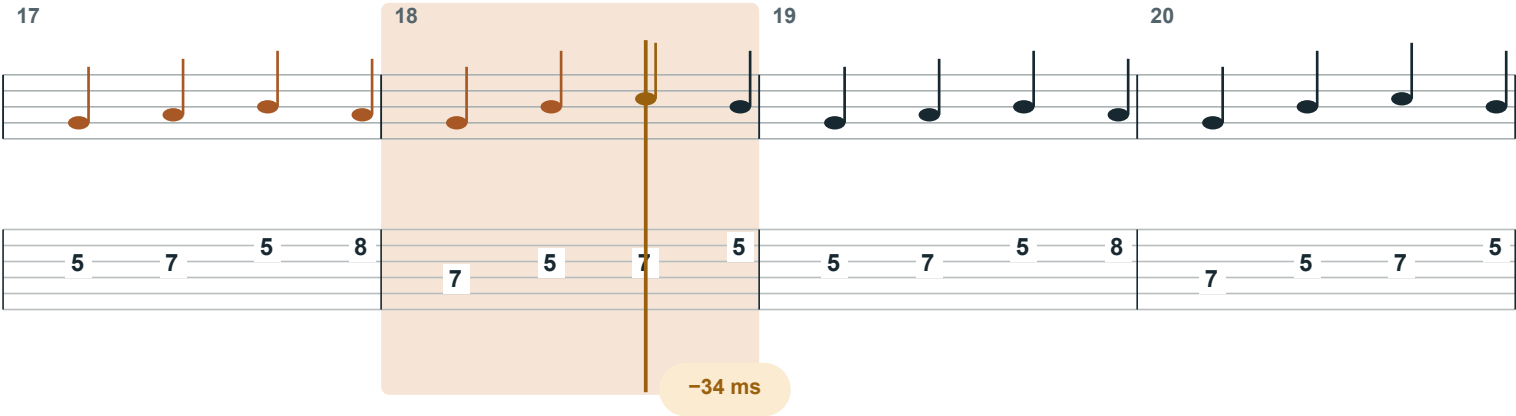
Your role: lead guitar · Band: drums + bass + keys · Original practice sketch

Score + TAB

72 BPM / 90 target

Loop 17–20

17181920



5758757557587575

ListenPlay alongCheck take

Pause

-Tempo+

A [ 17 ] B [ 20 ]

Count-in: 1 barRest: 2 beatsRepeat: 3

Band challenge

Stay with the pulse when the click drops out.

The band keeps a steady reference.

THIS TAKE

Listen for the change

Detailed feedback appears when the loop ends.

BAND MIX

Your part muted

Bass70%Drums55%

Keys35%Click20%

1 / A musical set

Move between the imported piece and a related jam without resetting input, instrument or mixer.

2 / A shared form

Notation / TAB, loop handles and tempo remain visible; a chord strip can replace TAB in open jams.

3 / Between-take coaching

The band does not chase the student during an assessed pass. Adaptations happen at the next chorus.

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04 / 17

# Observe → explain → practice → return

Useful feedback needs a visible cause, a manageable next action and a check that the action helped.

Pocket

LibraryPracticeJamProgress

Input: ready

## The groove held. The entry rushed.

Bars 17–20 · 72 BPM · Take 3

28 / 31  
assessed notes matched

26 ms  
median [timing]

31 / 32  
notes assessable

17 18 19 20

5 7 5 8 7 5 7 5 5 7 5 8 7 5 7 5

• Matched ▲ Early × Pitch mismatch ? Unclear

ONE NEXT STEP

You came back early after the quiet bar.  
Try the same entry again.

Two bars with a click, one without. Repeat the entry at 64 BPM, then bring the full band back for a written phrase check.

Try the gap challenge

Replay evidence · Mark incorrect

### How the coach chooses this exercise

A timing drift selects a metronome-gap task; unstable changes select roots-only or two-chord alternation; phrase length issues select call-and-response. Recheck with the original band and score afterward.

### When the recording is unclear

Show “Could not assess this note.” Offer a level or input check; preserve the take without changing mastery.

# Reward evidence of learning

Collect rehearsal cards for pocket, consistency and recall, each tied to a role, groove and tempo. Personal bests compare the same form and backing seed; a nice-sounding saved take is itself a reward.

PocketLibraryPracticeJamProgressInput: ready

## A set you can play with confidence.

Same phrase, same rubric. Your last 4 sessions.

GROOVE BY CHORUS / 72 BPM

Chorus 1

Re-entry: 44 ms early

Chorus 2

Re-entry: 28 ms early

Chorus 3

Re-entry: 14 ms early

Saved: shuffle in A / bass role

Groove / shuffle at 72 BPM

Three consistent choruses under this challenge rule.

Piece / phrase 04 at 72 BPM

Exact-score accuracy stays separate from jam results.

Next: a full rehearsal

Connect the entry with the following four bars.

## The reward rule

Written parts use exact-score criteria. Constrained jams use the announced task. Free improvisation offers timing and phrase observations without a pitch-accuracy grade or a musical-quality score.

## Player control

Choose free jam, a challenge or an exact part. Accept a suggested variation at the next chorus, lock the arrangement, or keep playing with feedback hidden.

## A reason to return

Offer one familiar groove, one weak transition and one optional variation. A saved set makes returning easy; new sounds never gate core practice tools.

## Keep comparisons honest

Store tempo, phrase, tuning, rubric, input quality and arrangement version. Show unknown separately; compare like with like.

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06 / 17

# A plain-language request becomes an editable band

“Shuffle beat ii–V–I in A” is ambiguous. Show the chosen harmony and feel before playback; users can edit either.

PocketLibraryPracticeJamProgressInput: ready

### Build a band in a sentence

shuffle beat ii–V–I in A

Interpreted as A major · jazz sevenths · edit any chip

A major

4/4

90 BPM

Shuffle 2:1

8 bars

Jazz trio

Bm7

bars 1–2

E7

bars 3–4

Amaj7

bars 5–8

Drums: shuffle

Bass: walking

Keys: sparse comping

Your role: guitar · Mode: free improvisation · Feedback: groove observations only

Preview band

Start practice

Same score, loop and feedback controls

## Why it belongs here

Jam is the main creative surface. Arrange idiomatic bass, drums and comping from a small curated style library. Rehearsal changes happen on form boundaries, with an explicit preview.

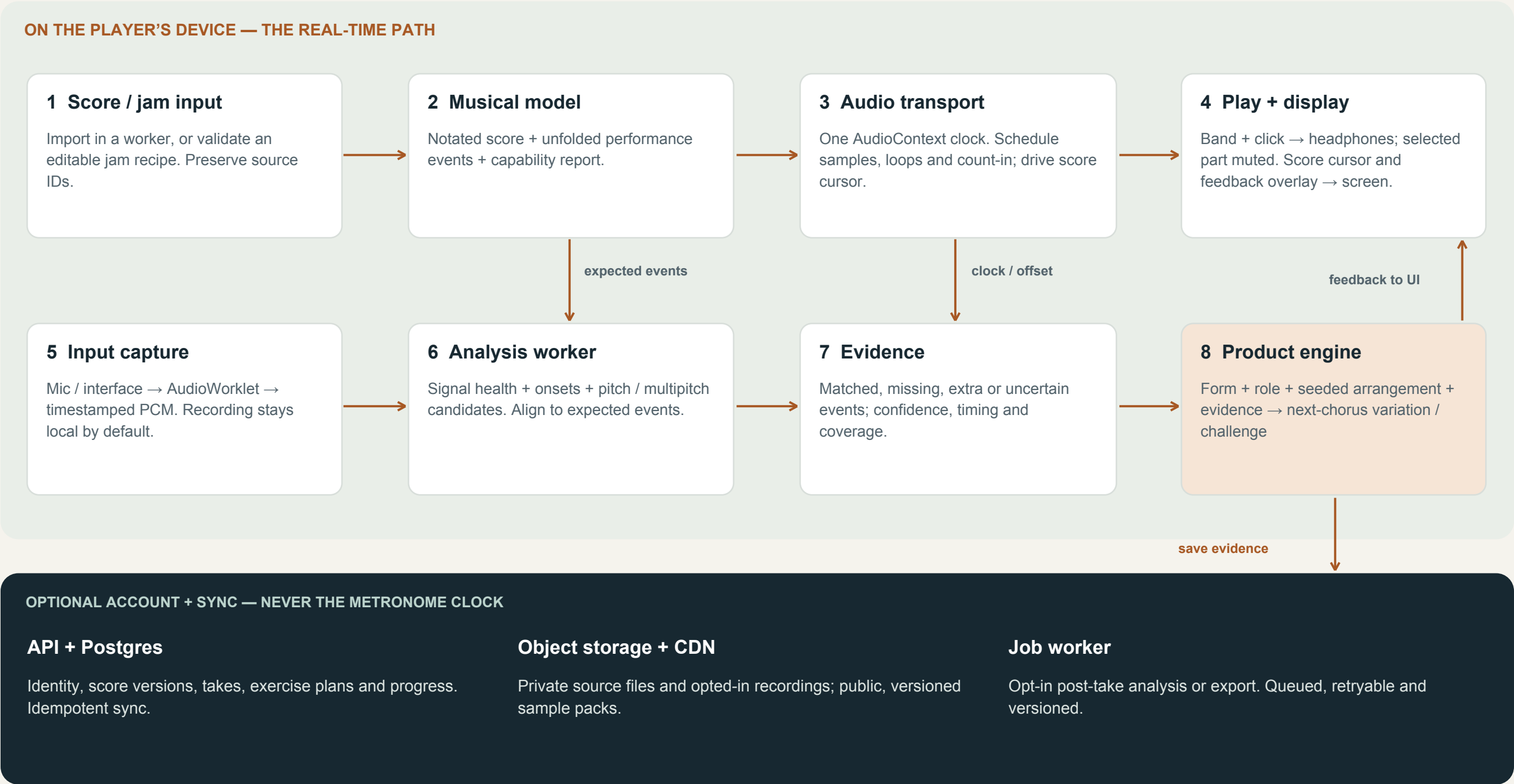
## Musical defaults

A major; Bm7 → E7 → Amaj7; 2 + 2 + 4 bars; 4/4 at 90 BPM; eighth-note shuffle 2:1. Offer triads, minor key or dominant blues explicitly.



# One musical timeline connects the whole system

Proposed stack: TypeScript + React/Vite; alphaTab adapter; Web Audio; worker-based analysis; a small API and database. [1–8]



# The supplied files already reveal the hard cases

Local metadata inspection with alphaTab 1.8.4, plus GP8 XML inspection. Parsing succeeded for all three; rendering and sound were not validated.

| Supplied file              | Observed structure                                                            | What it changes in the design                                                                                                            |
|----------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Born under a bad sign .gp4 | 6 tracks · 84 notated bars · 90 BPM<br>Bass, guitars, drums, piano and brass. | Start with bass and a selected guitar part. Track names and instrument metadata need user confirmation.                                  |
| Hotel California .gp4      | 10 tracks · 121 bars · 74 BPM<br>122 bent notes; 108 tie destinations.        | Large mixer and score layout; sustain and bends. Import emitted custom-bend fallback warnings; inspect those events.                     |
| Solo 3 – Comfort Food .gp  | GP8.1.2 · 1 track · 53 bars · 88 BPM<br>Embedded MP3; 32 sync points; swing.  | Separate notation time from recorded performance time. Blank title metadata needs a filename fallback; preserve tuplets and grace notes. |

## Import → validate → normalize

1. Check magic bytes, archive size, file count and decompression limits.
2. Parse off the UI thread; disable external XML references.
3. Confirm the playing track, instrument, tuning and capo.
4. Report unsupported notation, sound and grading separately.
5. Persist original bytes + parser version + immutable score version.

## A notation model and a performance model

Keep bars, voices, rational durations, rests, ties, ornaments, strings/frets, tuning, capo, written/sounding pitch and source IDs. Unfold repeats into occurrence IDs; map source notes to each occurrence.

Compile expressive timing and tempo into expected events. A tied continuation is not a new attack. Keep MIDI as an export of the richer score model.

# Treat the backing band as an instrument

Sample-based playback is part of the core product. Broad format support and beautiful tone are separate acceptance gates. [3–5, 9–10]

| Layer           | First complete release                                                                                   | Quality and expansion plan                                                                                                                                                                |
|-----------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Guitar + bass   | Curate or commission clean picked / fingered multisamples with 3+ velocity layers and alternate attacks. | Map tuning, key ranges, release and mute behavior. Add per-voice bends, slides and amp/cab effects; do not assume access to Guitar Pro RSE assets.                                        |
| Drums + comping | Audition a web-sized Virtuosity Drums subset and VCSL keys; retain source licenses.                      | Author dynamics, choke groups, bass voice leading and sparse comping. Commission missing core sounds with web redistribution rights.                                                      |
| Pack pipeline   | Offline tooling trims, normalizes and maps licensed WAVs into manifests, zones and compressed assets.    | Load active tracks first; decode before playback; cache versioned packs. Target a 15–40 MB starter download and ≤256 MB decoded samples on the baseline laptop; validate with real packs. |
| Fallback sounds | A permitted GM SoundFont covers unusual instruments; label it as standard sound quality.                 | Quality packs must pass listening tests before launch. The fallback is continuity during downloads, not the premium-sound acceptance criterion.                                           |

## Transport owns musical time

One AudioContext and event scheduler drive playback. Prototype alphaTab’s IExternalMediaHandler for cursor sync; retain a bounds-based cursor fallback. Tempo scales event time, not sample pitch. Cancel events on seek; handle sustain, count-in, loop tails and tied notes explicitly.

## Imported recordings have their own path

Offer Reference recording and Generated band modes. GP8 sync maps align audio to bars. Fixed recordings need pitch-preserving time-stretching and cannot mute a single part without stems. Evaluate a licensed DSP implementation; avoid two independent playback clocks.

# Generate musical events, then play them beautifully

A constrained arranger is an achievable first product. A language model may help interpret wording later; it never owns the timing clock or the grading target.

## VALIDATED JAM RECIPE

|                  |                              |
|------------------|------------------------------|
| Key / mode       | A / major                    |
| Harmony          | Bm7 × 2   E7 × 2   Amaj7 × 4 |
| Meter / tempo    | 4/4 / 90 BPM                 |
| Groove           | shuffle-eighths / ratio 2:1  |
| Band             | drums + bass + piano         |
| Player role      | guitar                       |
| Form / variation | 8 bars / seed 1042           |
| Assessment       | free, constrained or written |

## From words to an arrangement you can inspect

1. A grammar recognizes key, Roman numerals/chord names, feel, meter, tempo and bar counts. Unknown terms stay highlighted; the structured form always works.
2. Normalize chord spelling and mode; validate durations and instrument ranges. Show defaults rather than silently guessing.
3. A style template chooses drum cells, bass patterns and comping density. Constrained voice leading keeps chord shapes playable and bass movement idiomatic.
4. Emit note, articulation and dynamic events plus an editable score/chord view. Save the recipe, generator version and seed.

## Make the groove intentional

For straight eighths, an offbeat is at 1/2 beat; a 2:1 shuffle places it at 2/3. Swing only eligible pairs, not explicit tuplets or imported timing twice.

Drums use authored accents, alternate samples and choke groups. Bass can use roots, fifths and reviewed approach-note rules. Comping selects sparse chord voicings within an instrument range. Humanization is seeded and bounded; store the exact reference events.

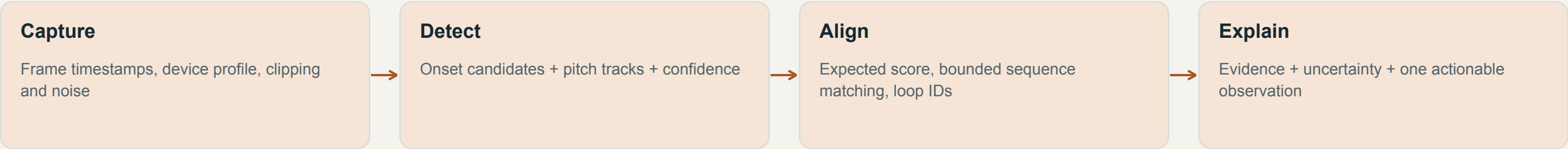
## Connect the band to practice

Mute the role the user will play, while retaining its expected events for a written or constrained task. In open improvisation, keep harmony visible and offer observations rather than a pitch grade.

Preview a 4–8-bar form, loop it, change key or tempo, and save it. Regenerate only at a stopped state or next form boundary. A generated arrangement enters the same transport, score, assessment and history pipeline as an imported piece.

# Know what was played before judging it

Proposed capability tiers, not claims of proven accuracy. Ship each assessed mode only after it passes an instrument- and input-specific benchmark. [5–8]



| Playing mode                                 | Assessment method                                                                                                                                                          | What the product promises                                                                                                                                   |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clean single notes<br>Initial grading mode   | Spectral-flux onsets and YIN-style pitch candidates; bounded sequence alignment with match, missing and extra events. Retain detected pitches before consulting the score. | Notes and timing on validated clean guitar / bass ranges. Target B0–E6, with longer windows for low bass; validate the range before advertising it.         |
| Chords and strumming<br>Post-take extension  | Multipitch model candidate (Basic Pitch) + score-conditioned event matching; group staggered string attacks.                                                               | Grade rhythm first; enable note-set checks only on validated chord families. Distortion, bleed and ambiguous voicings can remain ungraded.                  |
| Bends, slides, legato<br>Technique extension | Compare pitch contours and target arrival. Model ties and hammer-ons separately from picked onsets.                                                                        | Show expressive notes in notation and playback immediately. Add technique-specific grading after musician-labeled validation; audio cannot prove fingering. |
| Free improvisation<br>Jam mode               | Beat-relative onset and phrase observations; optional announced constraints such as roots on beat 1.                                                                       | No pitch-correctness score without a defined target. Musical taste, tone and creative choices are not reduced to a universal percentage.                    |

## LIVE AND FINAL FEEDBACK HAVE DIFFERENT JOBS

Live markers are provisional. Reconcile each take in a worker; preserve evidence and rubric versions. Keep prior results if models change.

# A score is only useful when its meaning is clear

Store raw counts, timing distributions, coverage and rubric version. The following thresholds are starting hypotheses for teacher validation.

### Separate pitch, timing and uncertainty

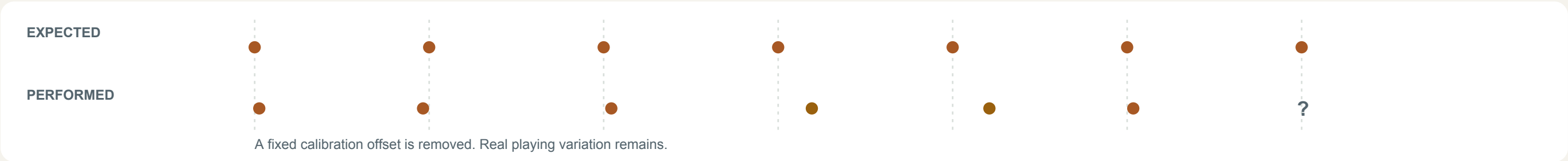
Pitch: correct expected notes / assessable expected notes; also show missing notes and extra attacks. Coverage: assessable expected notes / all eligible expected notes.

Timing: signed median, median absolute onset error and the 90th percentile on matched onsets. Preserve the bias: consistent lateness must not be automatically centered away.

### Keep the task stable during assessment

Bind every take to a track, phrase occurrence, tempo map, tuning, input profile, score version and rubric. Backing remains fixed while assessing; bounded alignment may recover the score location without forgiving actual timing errors.

Distinguish “missed” from “not assessable.” Silence on a healthy signal can be a miss; clipping or ambiguous pitch can be unknown.



### Calibrate the recording and playback path

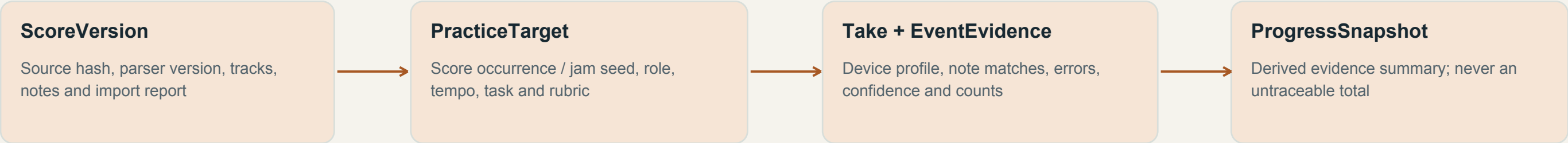
Measure round-trip delay with loopback where possible. A guided musical alignment is provisional because it includes human timing. Save uncertainty per device / sample rate; never recalibrate from each scored take. Recheck after route changes; unreliable Bluetooth timing disables fine timing grades.

### A proposed tempo ladder

For a simple picked-note drill: ≥90% note matches, ≥90% coverage, extras ≤5%, median |onset| ≤30 ms and p90 ≤60 ms. Require two qualified takes; offer +4 BPM next loop. Two difficult takes suggest −8 BPM or a smaller phrase. Retain only after a later check; adapt tolerances for expressive tasks.

# Save the evidence that makes progress reproducible

Start with a modular application and one backend. Keep sound scheduling and the initial feedback loop independent of network access.



### Band and challenge director

JamRecipe {chords, form, groove, role, seed, packVersion}; Challenge {mode, referenceEvents, criteria}; RehearsalCard {recipeVersion, takelds, result}.

POST /jam-recipes/validate; POST /arrangements; POST /challenges; GET /rehearsal-cards. Persist the arrangement seed for comparable takes.

### Local first, account optional

IndexedDB stores scores, takes and a sync outbox. UUIDs and idempotency keys prevent duplicate attempts. Immutable score and rubric versions avoid corrupting old progress. Optional account sync uses an authenticated API, Postgres and private object storage with short-lived URLs.

### API and job boundaries

POST /scores creates a version after upload validation; POST /takes records a completed attempt; GET /progress filters comparable evidence. A background job can refine an opted-in recording; results carry model and rubric versions.

Use a queue only for post-take work. Signed uploads, ownership checks, quotas and archive limits protect the shared backend.

### Privacy, recovery and release operations

No raw recording upload by default. Explicitly save or upload a take; offer per-take and account deletion plus export. Recordings are not model-training data without separate opt-in.

Show local / syncing / saved / failed states. Retry safely after offline use. Monitor input failure, analysis time, underruns and disputed feedback; exclude private score/audio content from telemetry.

# A gated path to a complete Pocket beta

Planning estimate: 22–28 weeks with three engineers, a half-time designer, fractional QA and a musician/teacher partner. Ranges are assumptions, not vendor estimates.

| Phase / duration                       | Primary owner                     | Deliverable and exit criterion                                                                                                                                                                                              |
|----------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 · Prove the risks<br>2 weeks         | Audio/DSP + frontend              | Render and audition the three supplied files; label input recordings; verify clock and external-cursor integration. Exit: fidelity report, sample rights inventory and a measurable assessment baseline.                    |
| 1 · Complete playback<br>3 weeks       | Frontend + audio                  | Import report, score/TAB, selected track, loop, speed, count-in and mixer. Canonical events handle repeats and tempo changes. Exit: deterministic playback and cursor tests; seek and loop sustain behavior checked.        |
| 2 · Hear the student<br>4–6 weeks      | DSP + platform + sound contractor | Capture, device checks, alignment, monophonic evaluation and evidence review. In parallel, curate quality sounds. Exit: precision, coverage, latency and blind-listening gates pass on supported setups.                    |
| 3 · Deliver the concept<br>8–10 weeks  | Product + full team               | Build the typed jam composer, three authored styles, role-aware arrangements, challenge director and rehearsal cards. Exit: musicians approve the groove and can tell exactly what the challenge evaluates.                 |
| 4 · Finish the whole loop<br>3–4 weeks | Full stack + product              | Add or complete jam recipes, arrangement-to-score, account sync, progress, retention checks and recovery states. GP8 reference audio uses a validated sync/time-stretch path. Exit: full user journey works across reloads. |
| 5 · Pilot and harden<br>2–3 weeks      | Full team + teacher + QA          | Run the musician pilot; test Safari/Chromium, long pieces and input changes. Audit accessibility, storage, sample licenses and feedback disputes. Exit: launch gates met and supported modes published.                     |

## After the complete beta

Budget 10–16 more weeks for validated chord families, bend/slide assessment and more styles; research may extend this. New instruments require new profiles and benchmarks. Core note, timing, training, progress and jam workflows already exist in the beta.

## Cost drivers

Three engineers dominate cost. Allow \$10k–\$30k for sound/style work; obtain quotes after auditioning the first pack.

# Define the promises, then test them

All numeric gates below are proposed acceptance targets. None was measured on a finished application during this design exercise.

| Risk / requirement           | Evidence needed before release                                                                                                                                                                                                                                                                                                 |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Import and notation fidelity | Compare GP3/4/5/6/7/8 and MusicXML fixtures to trusted notation and MIDI references. Include tuning, capo, repeats, alternate endings, tempo/meter changes, ties, swing and tuplets. The three local files alone are insufficient.                                                                                             |
| Trustworthy feedback         | Collect consented clean DI and mic takes from guitarists and bassists, plus noisy/distorted negatives. Split by player and instrument. On supported cases: $\leq 2\%$ false mistake flags among teacher-labeled correct events and $\geq 90\%$ coverage and $\geq 85\%$ recall of labeled mistakes; report all three together. |
| Clock and response           | Test injected timing offsets and loopback recordings. Target $\leq 10$ ms p95 residual error on calibrated wired setups; onset markers $\leq 150$ ms p95, final review $\leq 5$ s for a 30 s mono take on a stated midrange laptop. Low bass pitch may appear later.                                                           |
| Sound and reliability        | Blind-listen with $\geq 12$ musicians; target $\geq 8$ preferring the quality band over fallback. Test 30-minute sessions, seeks and tempo changes without stuck notes or underruns on the baseline device. Preload and cache only the needed patches.                                                                         |
| Learning and usability       | Recruit 12 guitarists and bassists who use backing tracks. Blind-audition the band, then compare open jamming with one focused challenge and a return to the written piece.                                                                                                                                                    |
| Concept-specific failure     | Musicality and tone quality are the product. A technically correct but repetitive backing band will fail. Budget for arrangement authors and listening tests, not just a text-generation API.                                                                                                                                  |

## Release scope covers every requested workflow

Upload → score/TAB → sampled playback → tempo/loop → live audio → supported note/timing evidence → generated exercise → progress history. Typed jam request → editable form → sampled backing → the same practice and review tools.

## Broader validation

Keyboard controls, text alongside colors, large targets, reduced motion and a tablet score view. Phone gets a focused single-system view.

# What is verified, what is proposed, what comes next

External technical sources reviewed September 29, 2026. Links are clickable. Product behavior, thresholds, schedules and budgets are design proposals.

[1] **alphaTab format catalog**

Import candidates: GP3–8 and MusicXML. Verify each feature on a fixture corpus.

[2] **alphaTab GP8 compatibility**

Reading, engraving and audio support differ; importing a feature is not proof of fidelity.

[3] **alphaTab external audio synchronization**

External playback hooks and GP8 sync data. Built-in backing and synthesis are alternative modes.

[4] **alphaTab project / licensing**

MPL-2.0; notation, interactive playback and SoundFont support. Review shipped dependencies and notices.

[5] **W3C Web Audio specification**

AudioContext, AudioWorklet and output timestamps underpin the proposed timing pipeline.

[6] **W3C Media Capture and Streams**

Input permissions and audio constraints. Inspect actual settings; requests are not guarantees.

[7] **Spotify Basic Pitch repository**

Candidate for post-take multipitch analysis. Best with one instrument; benchmark before adopting.

[8] **Basic Pitch research paper**

Bittner et al., 2022. Research basis, not evidence of student-assessment accuracy.

[9] **Versilian Community Sample Library**

CC0 library candidate for selected accompaniment sounds; audition and curate per patch.

[10] **Virtuosity Drums source and license**

Candidate drum source. Keep the exact pack license and provenance with derivative assets.

LOCAL EVIDENCE

Source files: main checkout /guitarpro; metadata and import warnings recorded in research/file-analysis.json. Prototype reviewed: timingAnalyzer.ts, microphoneRecorder.ts, metronome.ts, sampler.ts and pitch.ts. Existing onset logic and UI ideas are references; this plan does not depend on keeping their implementation.

DECISION THIS DESIGN SUPPORTS

Most distinctive jam experience and strongest immediate musical appeal. Highest arrangement and sample-production effort; avoid starting with an unrestricted AI band.